

Luis López Díaz

Ottawa, Canada

📞 (+1) 343 2048374 • ✉ lulopdz@gmail.com • 🌐 lulopdz.github.io

*“Everyone you will ever meet knows something you don’t”
- Bill Nye*

As a **Ph.D. candidate** at Carleton University (Canada), I am actively researching for a sustainable future power system. I am a proud member of the [Alternative Pathways for the Energy Transition \(APEX\)](#) research group. My research interests focus on Power and Energy Systems Planning. I am interested in mathematical models that address decision making under uncertainty with applications to modern power and energy systems. I hold a Master’s degree in Electrical Engineering (2020) from Universidad del Norte, Colombia. My academic career have been complemented by industry experience through engaging in consulting projects.

Education

- Phd Student**
Carleton University
Supervisor: Dr. Kristen Schell

Sep 2023 – Now
Canada
- Phd in Engineering Systems***
Skolkovo Institute of Science and Technology
Supervisor: David Pozo Camara Ph.D.
**Note: Discontinued due to unforeseen geopolitical circumstances.*

Jan 2021 – Apr 2022
Russia
- Visiting Student**
Skolkovo Institute of Science and Technology
Supervisor: David Pozo Camara Ph.D.
Co-supervisor: Álvaro González Castellanos Ph.D.

Sep 2019 – Dec 2019
Russia
- Master of Science in Electrical Engineering**
Universidad del Norte
GPA: 4.68/5.0
Thesis: Optimal Location and Sizing of Distributed Generation with non-Conventional Renewable Energies
Supervisor: Ingrid Oliveros Pantoja Ph.D.
Relevant Courses: Stochastic Processes, Optimization, Electrical Planning, Smart Grids, Electricity Markets.

2018 – 2019
Colombia
- Study Trip**
DAAD Program
Universities: JGU Mainz, Max Planck Institute for Polymer Research, Frankfurt University of Applied Science, Universitat Bayreuth, FAU Erlangen.
Presentation: Renewable Energies in Colombia: Technologies and Current Laws.

Oct 2018
Germany
- Bachelor in Electrical Engineering (double degree)**
Universidad del Norte
GPA: 4.49/5.0
Thesis: Voltage Variations Prediction in Substations
Advisor: Javier Guerrero Sedeño Ph.D.

2014 – 2016
Colombia

- **Bachelor in Electronics Engineering** **2011 – 2015**
Universidad del Norte *Colombia*
GPA: 4.46/5.0
Thesis: Software of Rendering and Visualization of Medical Images of the Heart
Advisor: Juan Tello Portillo M.Sc.

Work Experience

 **Academia:** 6+ years |  **Industry:** 2+ years

- **Carleton University**  -
Research/Teaching Assistant Sep 2023 - Now
 - Assisting in course preparation
 - Assist in energy transition research projects
- **Global EdTech Company**  **7 months**
Lead Academic Mentor Nov 2022 – Jun 2023
 - Tutor Assessment and Quality Assurance
 - Session Evaluations, Calibrations, Assessments and Mentoring
 - Development of Teaching and Learning Resources
- **Consultorías y Servicios DR**  **10 months**
Power systems engineer Jan 2022 – Nov 2022
 - Conducted Power System Simulations
 - Verified Compliance of New Generation Installations
 - Worked closely with regulatory bodies and stakeholders
- **Universidad del Norte**  **3+ years**
Adjunct Lecturer Jul 2022 – May 2023
 - Course 2023-01: IEL1021–Circuits Analysis II.
 - Course 2022-02: IEL1021–Circuits Analysis II.*Adjunct Lecturer* Jan 2020 – Dec 2020
 - Course 2020-02: IEL1021–Electrical Machines I.
 - Course 2020-01: IEL1021–Circuits Analysis II.*Graduate Teaching Assistant* Jan 2018 – May 2019
 - Course 2019-01: IEL1021–Circuits Analysis II.
 - Course 2018-02: IEL8435–Instrumentation and Measurements.
 - Course 2018-01: IEN4120–Digital Electronics.*Teaching Assistant* Jul 2016 – Nov 2016
 - Course 2016-02: IEL1021–Circuits Analysis II.
- **Universidad Pedagógica y Tecnológica de Colombia**  **1+ Month**
Visitor Lecturer Aug 2021 – Oct 2021
 - Course: ELECIII: Renewable energies.

- **Institución Universitaria ITSA**  **2+ years**
Adjunct Lecturer Jan 2020 – Dec 2021
 · Course: ELC31–Residential electrical installation.

Adjunct Lecturer Jan 2019 – Aug 2020
 · Course 2019-02: IMD02–Integrative project II.
 · Course 2019-01: ELE10—Electrical circuits.
- **Unión Temporal Iluminaciones Zona Bananera**  **1+ years**
Street Lighting Engineer Nov 2016 – Jan 2018
 · Development of brief, design, specification, preparation of contract documents, procurement, supervision and project management of Electrical Engineering projects.

Projects

 **Research:** 5+ years |  **Consulting:** 1+ years

1. **Graduate Student Researcher**  **Jul 2023 - Now**
CarletonU-ECCC: Bending the Curve with NETs Canada
 Project: NETS 2050: Developing a new modelling framework for negative emissions technologies [\[link\]](#)
Project Leader: Dr. Kristen R. Schell
2. **Graduate Student Researcher**  **Jan 2021 – Feb 2022**
MIT-Skoltech Next Generation Program Russia
 Project: Market Design for Electromobility: New Formulations, Models and Algorithms [\[link\]](#)
Project Leader: David Pozo Camara Ph.D.
3. **Street Lighting Engineer**  **May 2020 – Jun 2020**
Inmobiliaria e inversiones ISAN S.A.S Colombia
 Project: Provide Electrical Engineering consultancy services to internal and external clients in relation to Street Lighting.
Project Manager: Adm. Orieta Rodríguez Reales.
4. **Electrical Project Engineer**  **Feb 2020**
Universidad del Norte Colombia
 Project: Grid connection study of a PV plant 60 MW @ 110kV. Impact studies to determine the feasibility of generator connection to the distribution network.
Project Manager: Ingrid Oliveros Pantoja Ph.D.
5. **Graduate Student Researcher**  **Feb 2018 – Feb 2020**
Universidad del Norte Colombia
 Project: Development of a hardware/software platform for modeling the integration of small renewable distributed generation plants in low-voltage electricity grids in the Caribbean Region to evaluate their impact on the system stability and the energy balance of the network.
Project Manager: José Soto Ortiz M.Sc.
6. **Electrical Project Engineer**  **Dec 2018**
Universidad del Norte Colombia
 Project: Grid connection study of a gas thermal generator 50MW @ 110kV. Impact studies to determine the feasibility of generator connection to the distribution network..
Project Manager: Ingrid Oliveros Pantoja Ph.D.

- 7. Electrical Project Engineer** 

Universidad del Norte

Project: Design of a hybrid microgrid of renewable and non-renewable sources of electricity generation and distribution as a platform for research and technological development of solutions for non-interconnected environments.

Project Manager: Ingrid Oliveros Pantoja Ph.D.

Feb 2018 – Sep 2018

Colombia
- 8. Learning Content Designer**

Universidad del Norte

Project: Middle education promotion program developed in association agreement No. 006-2016 signed between the Universidad del Norte Foundation and the Special, Industrial, and Port District of Barranquilla.

Project Manager: Ingrid Oliveros Pantoja Ph.D.

Jun 2016 – Nov 2016

Colombia

Publications

Google Scholar: [\[link\]](#) | ResearchGate: [\[link\]](#)

- 2022.....
- [1] A. Gonzalez-Castellanos, L. Lopez, and D. Pozo, "On the Practical Use of Generalized Adaptive Partition Methods: Application to the Stochastic Unit Commitment Problem," in *2022 International Conference on Smart Energy Systems and Technologies (SEST)*, IEEE, Sep. 2022, pp. 1–6, ISBN: 978-1-6654-0557-7. DOI: [10.1109/SEST53650.2022.9898152](#).
 - [2] L. Lopez, A. Gonzalez-Castellanos, and D. Pozo, "Construction of Multi-period TSO-DSO Flexibility Regions," in *2022 International Conference on Smart Energy Systems and Technologies (SEST)*, IEEE, Sep. 2022, pp. 1–6, ISBN: 978-1-6654-0557-7. DOI: [10.1109/SEST53650.2022.9898423](#).
- 2021.....
- [3] A. Arango-Manrique, L. Lopez, J. Ramirez-Ortiz, and I. Oliveros, "Data-Driven Load Modeling to Analyze the Frequency of System Including Demand Response: A Colombian Study Case," *IEEE Access*, vol. 9, pp. 50 332–50 343, 2021, ISSN: 2169-3536. DOI: [10.1109/ACCESS.2021.3069006](#).
 - [4] A. Gonzalez-Castellanos, D. Pozo, S. Martínez, L. López, and I. Oliveros, "Economic Impact of Wind Generation Penetration in the Colombian Electricity Market," in *The 12th Mediterranean Conference on Power Generation, Transmission, Distribution and Energy Conversion (MEDPOWER 2020)*, Institution of Engineering and Technology, 2021, pp. 410–415, ISBN: 978-1-83953-524-6. DOI: [10.1049/icp.2021.1272](#).
 - [5] L. Lopez, A. Gonzalez-Castellanos, and D. Pozo, "Empirical Analysis of the Optimal Capacity Investment Solutions in Distribution Grids," in *2021 IEEE Madrid PowerTech*, IEEE, Jun. 2021, pp. 1–6, ISBN: 978-1-6654-3597-0. DOI: [10.1109/PowerTech46648.2021.9494965](#).
 - [6] L. Lopez, A. Gonzalez-Castellanos, D. Pozo, M. Roozbehani, and M. Dahleh, "QuickFlex: a Fast Algorithm for Flexible Region Construction for the TSO-DSO Coordination," in *2021 International Conference on Smart Energy Systems and Technologies (SEST)*, IEEE, Sep. 2021, pp. 1–6, ISBN: 978-1-7281-7660-4. DOI: [10.1109/SEST50973.2021.9543349](#).

2020

- [7] J. D. Acosta Dangond, I. Oliveros Pantoja, J. D. Soto, L. Ripoll Solano, and L. Lopez, "A Review of Use of non-Conventional Renewable Energies in Healthcare Facilities and Hospitals," in *2020 IEEE ANDESCON*, IEEE, Oct. 2020, pp. 1–6, ISBN: 978-1-7281-9365-6. DOI: [10.1109/ANDESCON50619.2020.9272060](https://doi.org/10.1109/ANDESCON50619.2020.9272060).
- [8] L. Lopez, I. Oliveros, A. Gonzalez-Castellanos, and D. Pozo, "Distributed Generation Allocation Under Uncertainty," in *2020 IEEE Power & Energy Society General Meeting (PESGM)*, IEEE, Aug. 2020, pp. 1–5, ISBN: 978-1-7281-5508-1. DOI: [10.1109/PESGM41954.2020.9281997](https://doi.org/10.1109/PESGM41954.2020.9281997).
- [9] L. Lopez, I. Oliveros, L. Torres, L. Ripoll, J. Soto, G. Salazar, and S. Cantillo, "Prediction of Wind Speed Using Hybrid Techniques," *Energies*, vol. 13, no. 23, p. 6284, Nov. 2020, ISSN: 1996-1073. DOI: [10.3390/en13236284](https://doi.org/10.3390/en13236284).
- [10] C. Restrepo, L. Lopez, and A. Arango-Manrique, "Evaluating the Effect in Frequency and Loadabilities Including Demand Response in a Power System: Colombian case," in *2020 IEEE International Conference on Environment and Electrical Engineering and 2020 IEEE Industrial and Commercial Power Systems Europe (EEEIC / I&CPS Europe)*, IEEE, Jun. 2020, pp. 1–6, ISBN: 978-1-7281-7455-6. DOI: [10.1109/EEEIC/ICPSEurope49358.2020.9160836](https://doi.org/10.1109/EEEIC/ICPSEurope49358.2020.9160836).
- [11] J. D. Soto, I. Oliveros, L. Ripoll, L. Lopez, and A. Rueda, "Pedagogical innovation to develop engineering skills in a power systems course through laboratory practices in a microgrid - UNIGRID," Valencia, Spain, Mar. 2020, pp. 3159–3163. DOI: [10.21125/inted.2020.0913](https://doi.org/10.21125/inted.2020.0913).

2019

- [12] L. Lopez, L. Afanador, L. Martinez, E. Jiménez, I. Oliveros, R. Castillo, J. Soto, M. Calle, and L. Ripoll, "Sizing of renewable energy sources to support resilience in distribution networks," *Procedia Computer Science*, vol. 155, pp. 535–542, Jan. 2019, ISSN: 1877-0509. DOI: [10.1016/J.PROCS.2019.08.074](https://doi.org/10.1016/J.PROCS.2019.08.074).
- [13] L. Lopez, J. Doria-Garcia, C. Pimienta, and A. Arango-Manrique, "Distributed Generation Allocation and Sizing: A Comparison of Metaheuristics Techniques," in *2019 IEEE International Conference on Environment and Electrical Engineering and 2019 IEEE Industrial and Commercial Power Systems Europe (EEEIC / I&CPS Europe)*, IEEE, Jun. 2019, pp. 1–6, ISBN: 978-1-7281-0653-3. DOI: [10.1109/EEEIC.2019.8783934](https://doi.org/10.1109/EEEIC.2019.8783934).
- [14] L. Lopez, J. Ramirez-Ortiz, A. Arango-Manrique, and I. Oliveros, "Impact on Voltage Profile of Demand Response in Residential Loads: Colombian case," in *2019 FISE-IEEE/CIGRE Conference - Living the energy Transition (FISE/CIGRE)*, IEEE, Dec. 2019, pp. 1–6, ISBN: 978-1-7281-4230-2. DOI: [10.1109/FISECIGRE48012.2019.8985011](https://doi.org/10.1109/FISECIGRE48012.2019.8985011).

2018

- [15] L. Lopez Diaz, A. Contreras Olmos, A. Cadena, W. S. Percybrooks, and J. P. Tello P., "Three-Dimensional Reconstruction of Cardiac Structures Using CT Images," in *Proceedings of the 2018 3rd International Conference on Biomedical Imaging, Signal Processing*, New York, NY, USA: ACM, Oct. 2018, pp. 74–79, ISBN: 9781450364775. DOI: [10.1145/3288200.3288210](https://doi.org/10.1145/3288200.3288210).

Honors and Awards

◦ Cum Laude (M.Sc. Thesis)

Universidad del Norte

Master of Science in Electrical Engineering

Thesis: Optimal Location and Sizing of Distributed Generation with non-Conventional Renewable Energies

2020

Colombia

- **Silver Medal**
Universidad del Norte
 Bachelor in Electrical Engineering
 The silver medal is awarded to the student with the highest GPA in their graduating class.

2016
Colombia
- **Cum Laude (B.Sc. Thesis)**
Universidad del Norte
 Bachelor in Electronics Engineering
Thesis: Software of Rendering and Visualization of Medical Images of the Heart

2016
Colombia
- **Graduate of Excellence**
Universidad del Norte
 Bachelor in Electronics Engineering

2015
Colombia
- **Dean's List**
Universidad del Norte
 List of the engineering department with the best GPA

2014
Colombia
- **Scholarship: Roble Amarillo**
Universidad del Norte
 Scholarship for bachelor studies at Universidad del Norte

2011
Colombia
- **Andrés Bello Award**
Ministerio de Educacion
 Best ICFES in Philosophy and Best ICFES in Language by territorial entity
 Resolution 987 of 2011

2010
Colombia

Supervision

- **Flórez Herazo, Nicolás Eduardo. Rueda León, Andrés Felipe.** (co-supervised with José Soto), 2020-01. Mobile stand equipped with non-conventional energy sources "NCES". [[link](#)]
- **Coba Pinedo, Moisés Alejandro. Suárez Gutiérrez, Juan David.** (co-supervised with José Soto), 2019-02. Software for the optimal determination of renewable generation sources in distribution networks. [[link](#)]
- **Brito Pinedo, Ian José. Calderón Yerena, Miguel Ángel.** (co-supervised with Rafael Castillo). 2019-01. Software for the evaluation of the effect of the solar and wind generation on distribution networks. [[link](#)]
- **Correa Pereira, Kennent.** 2018-02. (co-supervised with Rafael Castillo). Computer tool for dimensioning and locating nodes for generation distributed with non-conventional sources. [[link](#)]

Personal Skills

Language Skills.....

- **English:** C1, IELTS band score: 7.0.
- **Spanish:** Mother Tongue.

Computer Skills.....

- **Languages:** Julia (JuMP, Clustering), MATLAB, Python (scikit-learn, numpy, pandas)

- **Simulation software:** Simulink, DlgSILENT, PowerWorld
- **Other Tools:** Web Scraping (HTML, CSS), Git, LaTeX

Short Courses and Training

- Digital Learning Environments. IU Digital de Antioquia. *May 2020.*
- 3rd international conference on renewable energy: Energy Transition, a step towards the future. SER Colombia and World Energy Council Colombia. *Apr 2019.*
- Second Caribbean Regional Congress of Home Public Services. Lambda Foundation. *Sep 2016.*
- VIII International Seminar on Biomedical Engineering. Universidad de los Andes. *Apr 2016.*
- HP Accredited Technical Associate. HP ExpertONE. *Nov 2014.*
- Identification and analysis of integrated circuits and logic gates. Servicio Nacional de Aprendizaje SENA. *Sep 2014.*
- Design and development of printed circuits. Servicio Nacional de Aprendizaje SENA. *Jul 2014.*
- Fundamentals in the spectrum. Agencia Nacional del Espectro (ANE). *Sep 2013.*